

Financial Transactions Spend Analysis & Fraud Risk Assessment

– End-to-end analytics solution in Excel & Power BI

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1.0 Project Objective

This project transforms a **large-scale financial transactions dataset** into an **end-to-end analytics solution** focused on **customer spending behavior and fraud monitoring**. Using **one year of transaction data (2019)** comprising over **1.16 million transactions** across approximately **2,000 customers, 6,000 payment cards, and hundreds of merchant categories**, the project demonstrates the complete analytics lifecycle—from **data preprocessing and feature engineering to interactive dashboard development**.

The primary objective was to design and develop two portfolio-grade business intelligence dashboards that address distinct analytical use cases:

Spending Overview & Customer Segmentation: Understanding customer spending patterns, transaction behavior, demographic trends, and card usage characteristics.

Fraud Monitoring & Risk Analysis: Identifying fraud exposure, fraud-prone customer segments, transaction anomalies, and high-risk merchant categories.

The project showcases the combined use of Python for data preparation and feature engineering, Excel Power Pivot for multidimensional analysis, and Power BI for interactive dashboard design, data modeling, DAX-based calculations, and business storytelling.

2.0 Dataset Description

This project utilizes the **Financial Transactions Dataset** available on Kaggle which simulates banking transactions, customer profiles, payment card information, merchant categories, and fraud labels.

Dataset Link: [Financial Transaction Dataset \(Kaggle\)](#)

The original dataset consists of five source files:

1. **transactions_data.csv** - (13.3M rows, 12 columns) containing features such as `transaction_id`, `date`, `amount`, `client_id`, `merchant_id` etc.
2. **users_data.csv** - (2000 rows, 14 columns) containing features like `client_id`, `age`, `yearly_income`, `credit_score`, `total_debt` etc.
3. **cards_data.csv** - (6000 rows, 13 columns) containing features like `card_id`, `card_type`, `credit_limit`, `acct_open_date` etc.
4. **mcc_codes.json** - (109 rows, 2 columns) JSON file containing the features `mcc` and `mcc_description`.

5. **train_fraud_labels.json** – (8.9M rows, 2 columns) JSON file containing **transaction_id** and **is_fraud** features.

To make the analysis computationally efficient and focused on a single reporting period, the transaction dataset was filtered to include only transactions occurring during **calendar year 2019**, resulting in approximately **1.16 million transactions**.

Following data cleaning, preprocessing, and feature engineering in Python, the source files were transformed into a star schema consisting of one fact table and three dimension tables.

Final Analytical Data Model

Dataset	Records	Purpose
fact_transactions.csv	1.16M rows	Stores transaction-level measures and fraud indicators
dim_users.csv	2000 rows	Customer demographics, income, and credit characteristics
dim_cards.csv	6000 rows	Card attributes, credit limits, and card age metrics
dim_mcc.csv	109 rows	Merchant category descriptions

Key Engineered Features

Transaction Features

- Amount Bucket (Small, Medium, Large)
- Time of Day (Morning, Afternoon, Evening, Night)
- Fraud Flag (Fraud, Legitimate, Unknown)

Customer Features

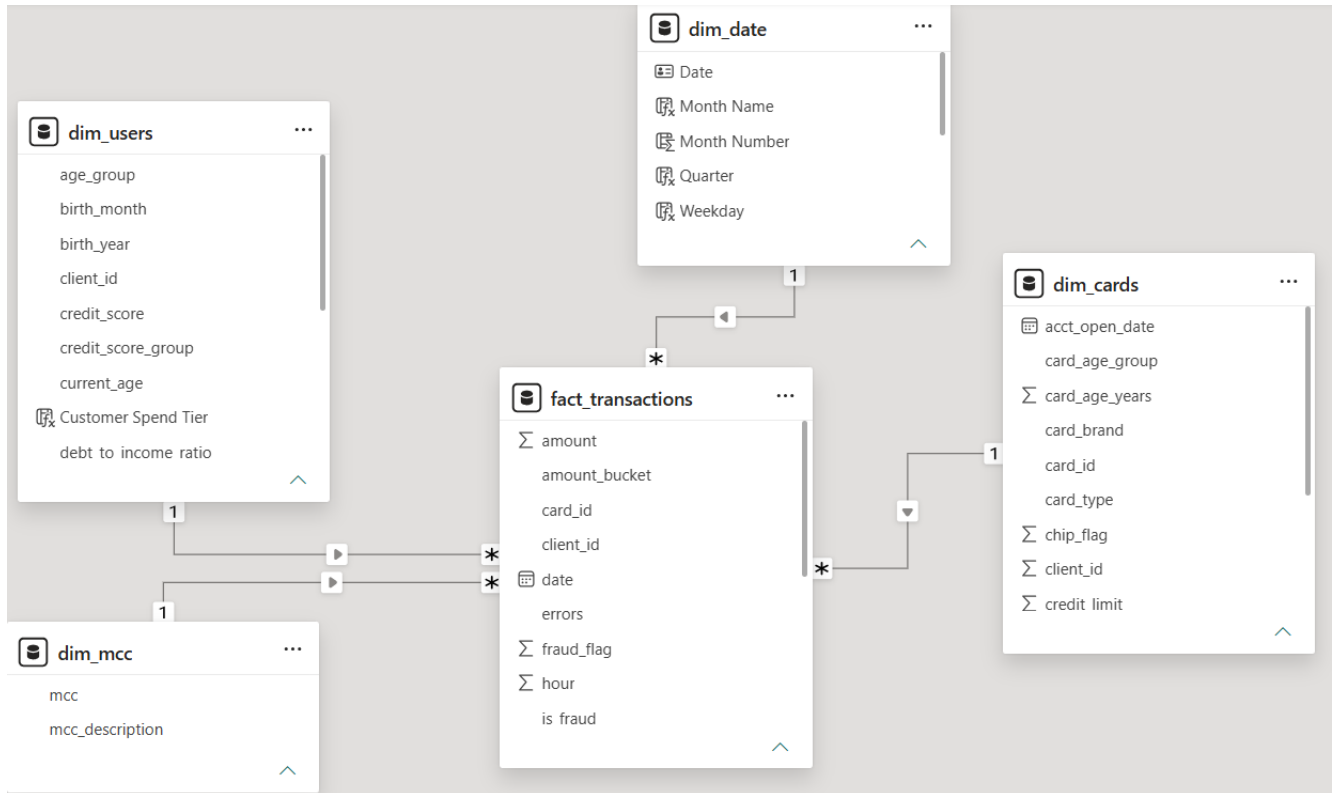
- Age Group
- Income Group
- Customer Spend Tier

Card Features

- Card Age (Years)

- Card Age Group

Star Schema Data Model



3.0 Python Data Preparation

The raw source datasets required substantial preprocessing before they could be used for analytical reporting and dashboard development. **Python (Pandas)** was utilized to perform **data extraction**, **preprocessing**, **cleaning**, **feature engineering**, and **dimensional modeling** to ensure **data quality**, **consistency**, and **analytical readiness**. The resulting datasets were **transformed** into a **star-schema structure** optimized for **Excel** and **Power BI** analysis.

The major data preparation activities are summarized below.

3.1 Data Preprocessing

- Processed approximately **13.3 million credit card transactions spanning 2010–2019** and filtered the dataset to **1.16 million transactions from 2019** to create a focused analysis period and reduce computational complexity.
- Extracted and transformed fraud labels and Merchant Category Code (MCC) mappings from JSON files into structured tabular datasets for downstream integration and analysis.
- Validated fraud label coverage by comparing transaction counts across datasets and confirmed that only **67% of all transactions had an associated fraud label**, indicating that the fraud dataset represented a subset of the overall transaction population.
- Integrated fraud labels into the transaction dataset using a **LEFT JOIN on transaction_id**, preserving all transaction records while retaining unlabeled transactions for separate treatment during fraud analysis.

3.2 Data Cleaning

- Standardized financial and date fields by converting currency-formatted strings and timestamps into appropriate numeric and datetime data types.
- Performed data quality checks to identify and resolve missing values, duplicate records, and data inconsistencies across all datasets.
- Validated negative transaction amounts and retained them as legitimate refund/reversal transactions rather than treating them as data errors.
- Investigated missing fraud labels and classified unlabeled transactions as **"Unknown"** instead of **"Legitimate"** to avoid introducing bias into fraud analysis.
- Removed non-analytical and sensitive attributes such as card numbers, CVV values, and customer addresses to improve data usability and model efficiency.
- Standardized categorical fields and validated dataset integrity prior to dimensional modeling and export.

3.3 Feature Engineering

- Created transaction-level features including transaction type, transaction hour, time-of-day segments, amount buckets, and fraud indicators to support spending and fraud analysis.

- Developed customer segmentation features such as age groups, income groups, credit score bands, debt-to-income ratios and customer value tiers for behavioral profiling.
- Engineered card-level attributes including card age, card age groups, expiry attributes, and card usage characteristics.
- Enhanced merchant analysis by integrating Merchant Category Code (MCC) descriptions into the analytical model.
- Prepared analysis-ready fact and dimension tables following a star-schema approach to support Excel analysis and Power BI dashboard development.

4.0 Excel Analysis and Power Pivot Modeling

Since the fact transactions dataset had 1.16M rows which are above the limit for an excel workbook the transaction dataset was aggregated in Python to create customer level data, monthly data and merchant level data. Also to perform statistical analysis like Hypothesis Testing a random sample of 100k transactions was sampled from the fact transactions dataset.

These four datasets were then imported into a simple excel workbook along with the dim_users dimension table to add customer level features wherever required using the XLOOKUP function in excel.

4.1 Excel Tools & Techniques Used

Excel

- Pivot Tables
- Pivot Charts
- XLOOKUP
- IFS/SUMIF/COUNTIF
- Data Validation
- Conditional Formatting & Heatmaps
- Dynamic Dashboards
- Risk Scoring Frameworks
- Percentile Ranking
- Data Analysis Toolpak

Power Pivot

- Data Model & Star Schema
- DAX Measures
- Calculated Fields
- KPI Metrics
- Interactive Slicers
- Matrix Analysis
- Cross-Segment Analysis

4.2 Fraud Monitoring Dashboard

Objective

Develop an operational fraud monitoring dashboard to identify merchant categories exhibiting elevated fraud exposure and prioritize investigation efforts through risk-based ranking.

Dashboard Image

Fraud Monitoring Dashboard						
Merchant Categories Monitored		40				
High Risk Categories		5				
Total Fraud Loss		\$116,403.59				
Average Fraud Rate		3.50%				
Highest Risk Category		Furniture, Home Furnishings, And Equipment Stores				
Merchant Category	Fraud Transactions	Fraud Loss	Average Fraud Amount	Fraud Rate	Weighted Risk Score	Risk Level
Music Stores - Musical Instruments	7	\$493.08	\$70.44	23.33%	0.67	Medium Risk
Computers, Computer Peripheral Equipment	40	\$3,946.74	\$98.67	14.87%	0.89	High Risk
Miscellaneous Fabricated Metal Products	3	\$221.44	\$73.81	11.54%	0.62	Medium Risk
Fabricated Structural Metal Products	5	\$2,300.11	\$460.02	10.87%	0.79	High Risk
Furniture, Home Furnishings, And Equipment Stores	36	\$16,493.11	\$458.14	9.76%	0.93	High Risk
Precious Stones And Metals	43	\$1,936.57	\$45.04	9.07%	0.75	High Risk
Miscellaneous Metal Fabrication	3	\$1,009.74	\$336.58	8.57%	0.68	Medium Risk
Floor Covering Stores	2	\$532.57	\$266.28	7.41%	0.58	Medium Risk
Bolt, Nut, Screw, Rivet Manufacturing	2	\$1,304.70	\$652.35	6.45%	0.68	Medium Risk
Steel Drums And Barrels	2	\$694.38	\$347.19	6.45%	0.60	Medium Risk
Gift, Card, Novelty Stores	11	\$19.53	\$1.78	5.53%	0.46	Medium Risk
Antique Shops	20	\$1,054.38	\$52.72	3.56%	0.61	Medium Risk
Coated And Laminated Products	1	\$725.85	\$725.85	2.94%	0.57	Medium Risk
Non-Precious Metal Services	24	\$2,779.20	\$115.80	2.41%	0.65	Medium Risk
Non-Ferrous Metal Foundries	21	\$3,237.16	\$154.15	2.27%	0.66	Medium Risk
Gardening Supplies	29	\$5,125.25	\$176.73	2.15%	0.73	High Risk
Industrial Equipment And Supplies	28	\$4,295.72	\$153.42	1.97%	0.68	Medium Risk
Semiconductors And Related Devices	25	\$2,865.29	\$114.61	1.86%	0.59	Medium Risk
Lodging - Hotels, Motels, Resorts	23	\$4,411.95	\$191.82	1.59%	0.66	Medium Risk
Passenger Railways	19	\$4,533.49	\$238.60	1.45%	0.66	Medium Risk
Family Clothing Stores	67	\$3,944.84	\$58.88	1.06%	0.58	Medium Risk
Women'S Ready-To-Wear Stores	7	\$649.70	\$92.81	0.79%	0.41	Medium Risk
Shoe Stores	9	\$545.61	\$60.62	0.73%	0.36	Low Risk
Department Stores	290	\$25,943.85	\$89.46	0.70%	0.65	Medium Risk
Florists Supplies, Nursery Stock And Flowers	4	\$62.13	\$15.53	0.51%	0.25	Low Risk
Discount Stores	101	\$5,542.33	\$54.87	0.47%	0.59	Medium Risk

Key Findings

- Merchant categories were evaluated using a weighted risk score combining fraud rate and total fraud loss, enabling balanced assessment of both fraud frequency and financial impact.
- Categories such as **Computers, Computer Peripheral Equipment, Furniture, Home Furnishings, and Equipment Stores**, and **Fabricated Structural Metal Products** exhibited the highest risk scores, driven by a combination of elevated fraud rates and significant fraud losses.
- While some categories generated substantial transaction volumes, their fraud rates remained relatively low, highlighting the importance of evaluating both frequency and monetary impact rather than relying on fraud counts alone.
- Categories such as **Artist Supply Stores, Craft Shops** and **Motor Freight Carriers and Trucking** demonstrated consistently low fraud rates and losses, resulting in the lowest overall risk classifications.
- The dashboard provides a structured framework for prioritizing fraud investigations, allowing analysts to focus on merchant categories with the greatest potential financial and operational risk.

4.2 Customer Risk Lookup Tool

Objective

Develop a self-service customer risk lookup tool that enables analysts to quickly retrieve customer demographics, spending behavior, credit characteristics, and fraud history using a dynamic customer ID selection interface.

Lookup Tool Image

Customer Lookup Table (2019)	
Select Customer ID	1163
Customer Profile	
Customer ID	1163
Income Group	High Income
Credit Score	489
Age Group	36-45
Transactions Metrics	
Total Spend	\$27,292.06
Total Transactions	397
Avg Spend	\$68.75
Fraud Metrics	
Fraud Transactions	8
Fraud Loss	\$860.99
Avg Fraud Amount	107.62
Fraud Risk	High Risk
Fraud Rate	2.02%

Key Findings

- The tool consolidates customer demographics, transaction activity, credit profile, and fraud metrics into a single view, eliminating the need to manually search across multiple datasets.
- Dynamic customer selection allows analysts to instantly retrieve customer-specific information, supporting rapid investigation and risk assessment.
- Conditional formatting highlights potential areas of concern such as low credit scores, elevated fraud losses, and high-risk customer classifications, enabling faster identification of customers requiring additional review.
- The integrated fraud metrics provide visibility into fraud exposure through measures such as fraud transaction count, fraud loss, fraud rate, and average fraud amount.

- By combining customer profile attributes with transactional and fraud indicators, the tool serves as an effective first-level investigation interface for customer risk monitoring and exception analysis.

4.3 Customer Spend Pareto Analysis

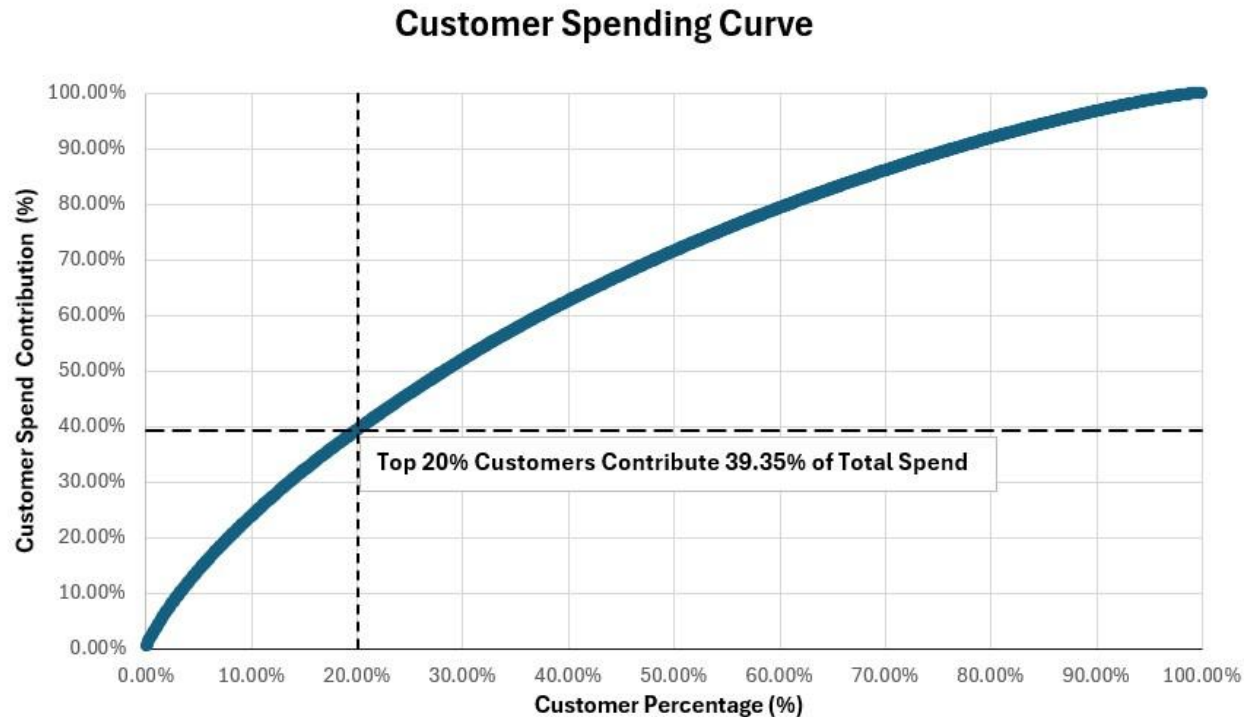
Objective

Analyze customer spending concentration to determine whether a small proportion of customers account for a disproportionately large share of transaction value and identify the contribution of top-spending customer segments.

Customer Spending Dashboard

Customer Spending Contribution Analysis						
Total Spend		\$49,475,055.31				
Total Customers		1207				
Top 10% Customers Spend		23.80%				
Top 20% Customer Spend		39.35%				
client_id ▾	total_spend ▾	customer_rank ▾	cumulative_spend ▾	cumulative_spend% ▾	customer % ▾	
96	\$193,921.09	1	\$193,921.09	0.39%	0.08%	
1686	\$178,566.16	2	\$372,487.25	0.75%	0.17%	
1340	\$166,219.34	3	\$538,706.59	1.09%	0.25%	
840	\$165,498.89	4	\$704,205.48	1.42%	0.33%	
464	\$159,181.66	5	\$863,387.14	1.75%	0.41%	
704	\$150,327.64	6	\$1,013,714.78	2.05%	0.50%	
1797	\$140,143.15	7	\$1,153,857.93	2.33%	0.58%	
989	\$138,806.94	8	\$1,292,664.87	2.61%	0.66%	
285	\$137,605.20	9	\$1,430,270.07	2.89%	0.75%	
488	\$135,366.04	10	\$1,565,636.11	3.16%	0.83%	
52	\$131,807.52	11	\$1,697,443.63	3.43%	0.91%	
1003	\$129,414.27	12	\$1,826,857.90	3.69%	0.99%	
490	\$127,601.35	13	\$1,954,459.25	3.95%	1.08%	
630	\$127,421.75	14	\$2,081,881.00	4.21%	1.16%	
1168	\$125,448.36	15	\$2,207,329.36	4.46%	1.24%	
1575	\$123,411.40	16	\$2,330,740.76	4.71%	1.33%	
1098	\$123,352.68	17	\$2,454,093.44	4.96%	1.41%	
1888	\$119,980.63	18	\$2,574,074.07	5.20%	1.49%	
1648	\$118,324.16	19	\$2,692,398.23	5.44%	1.57%	
1169	\$116,817.27	20	\$2,809,215.50	5.68%	1.66%	

Customer Pareto Curve Visualization



Key Findings

- Across 1,207 customers and **\$49.5M in total spend**, the top 10% of customers (≈ 121 customers) contributed **23.8%** of transaction value — roughly **\$11.8M** — while the top 20% (≈ 241 customers) accounted for **39.35%** ($\approx \$19.5M$), establishing a measurable but softer-than-classical concentration pattern.
- The **Lorenz curve shape** corroborates this: the gradual, convex accumulation without an inflection point indicates **spending power is spread relatively evenly across customer deciles**, rather than front-loaded among a small elite segment.
- **Single-customer concentration risk is minimal** — the highest-spending customer (Client 96, \$193,921) represents just **0.39% of total spend**, and the top 10 customers collectively account for only **3.16%** ($\approx \$1.57M$), meaning no individual or small cluster poses a material revenue dependency risk.
- From a **fraud-risk lens**, this spending diffusion means fraud losses are less likely to be concentrated in a handful of high-value accounts; instead, fraud exposure is spread across the portfolio, making broad transactional monitoring — rather than VIP-account surveillance — the more appropriate detection strategy.

4.4 Monthly Fraud Variance Tracker

Objective

Track month-over-month volatility in fraud losses, transaction volume, and fraud rate to quantify periods of elevated risk, distinguish systematic surges from isolated spikes, and provide a tiered early-warning framework that triggers investigative action based on predefined variance thresholds.

Monthly Variance Tracker Image

MONTHLY FRAUD VARIANCE TRACKER						
Total Fraud Loss			\$116,403.59			
Average Monthly Fraud Rate			0.12%			
Highest Fraud Loss Month			October			
Largest MoM Increase in Fraud Loss			110.18%			
Row Labels	Fraud Loss	Fraud Transactions	Fraud Rate	MoM % Change Fraud Count	MoM % Change Fraud Loss	Alert Flag
January	\$10,703.52	129	0.11%	N/A	N/A	N/A
February	\$8,398.68	113	0.11%	-12.40%	-21.53%	Normal
March	\$11,097.20	142	0.12%	25.66%	32.13%	Investigate
April	\$12,588.89	140	0.12%	-1.41%	13.44%	Monitor
May	\$17,129.41	168	0.14%	20.00%	36.07%	Investigate
June	\$10,475.60	132	0.11%	-21.43%	-38.84%	Normal
July	\$7,170.81	104	0.09%	-21.21%	-31.55%	Normal
August	\$13,121.19	165	0.14%	58.65%	82.98%	Investigate
September	\$8,291.47	90	0.08%	-45.45%	-36.81%	Normal
October	\$17,426.82	177	0.15%	96.67%	110.18%	Investigate

Fraud Loss Monthly Trend



Key Findings

- **Total fraud loss** over the observed period reached **\$116,403.59** across **1,360 fraudulent transactions**, with a **mean fraud rate of 0.12%**, masking significant intra-year volatility that a single annual figure would conceal.
- **October's MoM fraud loss surge of 110.18%**, the largest in the dataset — was accompanied by a **96.67% jump in fraud count**, indicating a coordinated or systemic fraud event rather than random variance.
- **August** exhibited the most disproportionate pattern: **an 82.98% MoM loss surge** against **only 58.65% count growth**, implying higher average loss per fraudulent transaction compared to other high-volume months.
- **September** acted as the strongest natural control point — **fraud count dropped 45.45%** and **fraud loss fell 36.81%** — confirming that fraud activity is not monotonically increasing and is likely tied to specific seasonal or transactional triggers.
- The Alert Flag framework correctly differentiates signal from noise: months flagged as *Investigate* (March, May, August, October) all show dual-axis pressure (both count and loss surging simultaneously), while *Monitor* (April) captures asymmetric risk where loss grows faster than count.

4.5 Hypothesis Testing

4.5.1 Do fraudulent transactions involve larger amounts than legitimate ones?

Business Question: Is the mean transaction amount significantly higher for fraudulent transactions compared to non-fraudulent ones?

Objective: To determine whether transaction amount can serve as a statistically reliable signal for distinguishing fraudulent activity — a finding that would directly inform threshold-based fraud alerting rules.

Method: Welch's Two-Sample t-test (unequal variances assumed), executed via Excel's Data Analysis ToolPak.

Null Hypothesis (H₀): The mean transaction amount is equal across fraudulent and non-fraudulent transactions.

$$\mu_{\text{fraud}} = \mu_{\text{nonfraud}}$$

Alternate Hypothesis (H_1): The mean transaction amount differs between the two groups.

$$\mu_{\text{fraud}} \neq \mu_{\text{nonfraud}}$$

Test Results

Mean of Legitimate transactions: \$ 96.35

Mean of Fraudulent transactions: \$ 42.81

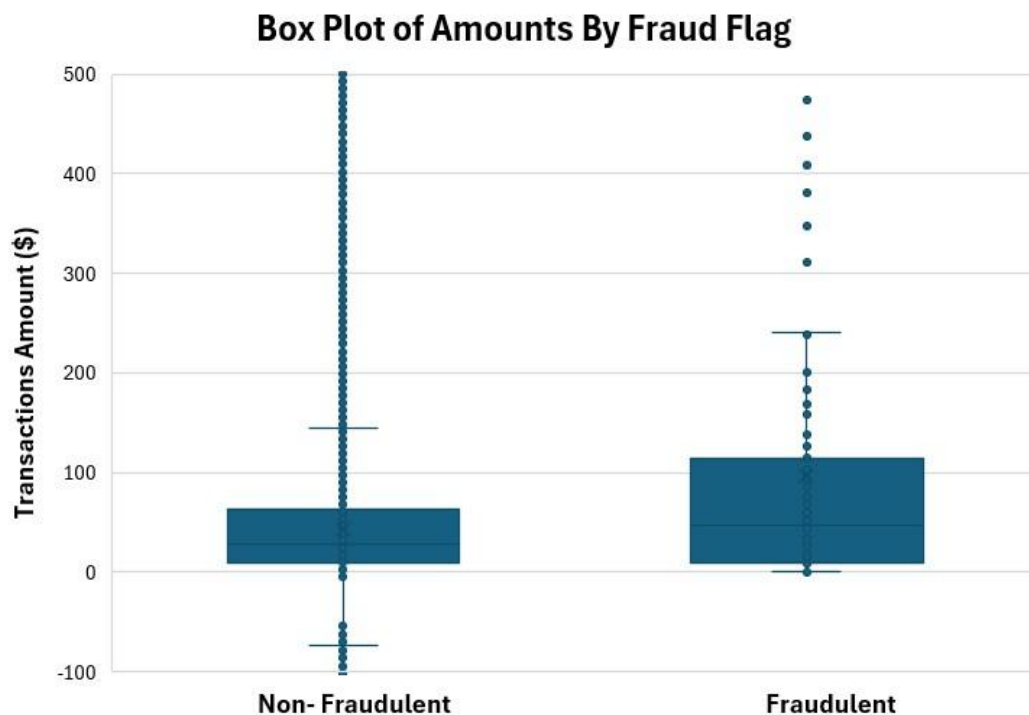
p-Value (two-tailed): 0.00039

Conclusion

With a p-value of 0.00039 — well below the 0.05 significance threshold — the null hypothesis is rejected. The difference in mean transaction amounts between fraudulent and non-fraudulent transactions is **statistically significant**, confirming that fraudulent transactions are, on average, more than **twice as large** as legitimate ones (\$96.35 vs. \$42.81).

This finding has a direct operational implication: **transaction amount is a viable input feature for fraud detection models**, and elevated transaction values should be treated as a risk signal warranting additional scrutiny in any rule-based or ML-based alerting framework.

Supporting Visualization — Box Plot of Transaction Amounts by Fraud Flag



The box plot provides strong visual corroboration of the test result. Several structural differences are immediately apparent:

- The **fraudulent IQR** (interquartile range) sits noticeably higher than the non-fraudulent IQR, confirming that the central mass of fraudulent transactions clusters at larger amounts.
- The **non-fraudulent distribution** is more tightly compressed with a lower median, consistent with the high volume of small routine purchases typical in legitimate transaction behavior.
- The **non-fraudulent box** extends into negative territory (likely representing returns or chargebacks), a pattern absent in the fraudulent group, which further differentiates the two distributions.

Taken together, the statistical test and the box plot converge on the same conclusion: **transaction amount is not randomly distributed across fraud status** — it is a meaningful, testable discriminator.

4.5.2 Does fraud rate differ across customer income groups?

Business Question: Is fraud occurrence statistically independent of a customer's income group, or does income segment meaningfully influence fraud likelihood?

Objective: To determine whether the observed variation in fraud rates across income groups — High, Low, Lower Middle, and Upper Middle — reflects a genuine association or is attributable to random sampling variation. A significant result would justify income-segmented fraud monitoring strategies.

Method: Pearson's Chi-Square Test of Independence (CHISQ.TEST Function in Excel), computed manually in Excel using observed transaction counts from a PivotTable and expected counts derived under the independence assumption.

Null Hypothesis (H_0): Fraud occurrence is independent of income group — the proportion of fraudulent transactions does not differ meaningfully across segments.

Alternate Hypothesis (H_1): Fraud occurrence is associated with income group — fraud rates vary significantly across at least one segment.

Test Results

Observed vs. Expected Fraud Counts by Income Group

Income Group	Observed Fraud	Expected Fraud	Chi-Square (Fraud Cell)
High Income	17	23.06	1.59
Low Income	33	20.13	8.23
Lower Middle	43	53.71	2.14
Upper Middle	31	27.10	0.56

p-Value: 0.00574

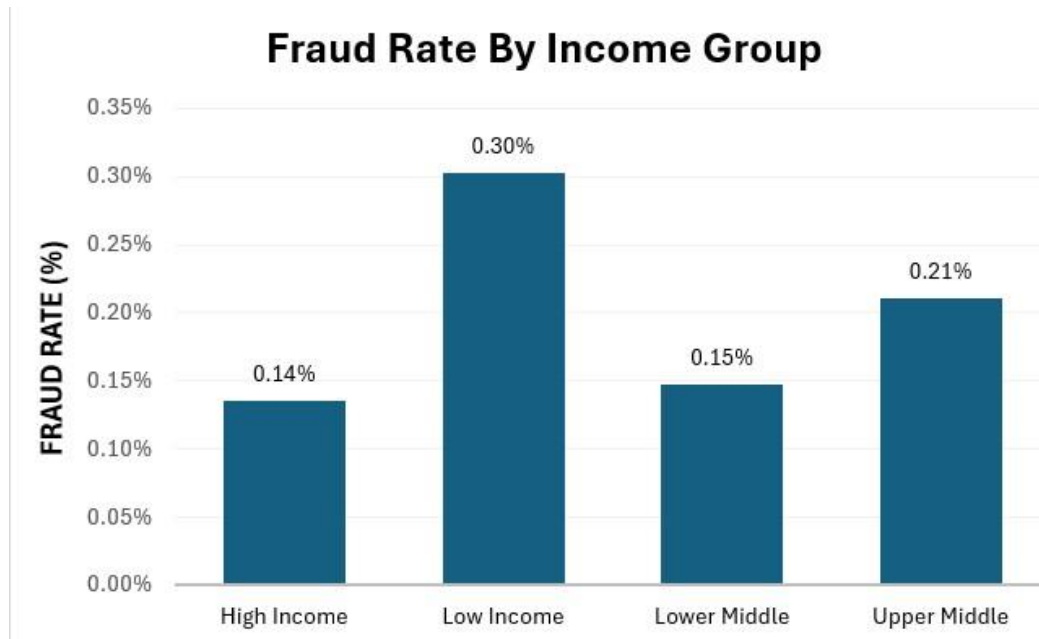
Conclusion

With a p-value of 0.00573 — below the 0.05 significance threshold — the null hypothesis is rejected. Fraud occurrence is **not independent of income group**; the distribution of fraud across segments deviates from what chance alone would produce, and this deviation is statistically significant.

The **Low Income group drives the result most forcefully**: its chi-square cell contribution of **8.23** is by far the largest across all cells, meaning observed fraud (33) substantially exceeded its expected count (20.13) — a **64% excess over expectation**. This is the single segment where the independence assumption breaks down most severely and where targeted fraud monitoring would yield the highest marginal return.

Conversely, **High Income customers were underrepresented in fraud** relative to expectation (17 observed vs. 23.06 expected), suggesting either stronger fraud controls, more cautious spending behavior, or lower targeting by fraudsters in that cohort.

Supporting Visualization — Fraud Rate by Income Group



The column chart translates the statistical finding into immediately interpretable terms. The **Low Income group's 0.30% fraud rate stands at more than double** that of High Income (0.14%) and Lower Middle (0.15%), visually anchoring the chi-square result. Upper Middle sits at an intermediate 0.21%, reinforcing that the relationship between income and fraud risk is not monotonic — it is the Low Income segment specifically, rather than a linear income gradient, that drives the significant association.

4.6 Power Pivot Analysis

While the Excel dashboards utilized aggregated datasets, Power Pivot was used to analyze the complete transaction dataset containing approximately 1.16 million records. The fact transaction table and supporting dimension tables (Customers, Cards, and Merchant Categories) were loaded into the Excel Data Model and connected using a star schema design.

The primary objective of this analysis was to leverage DAX measures and multidimensional filtering to identify relationships between customer demographics, credit characteristics, transaction behavior, and fraud exposure. This enabled fraud risk to be evaluated across different customer segments without requiring data aggregation prior to analysis.

The Following DAX measures were added to the model:

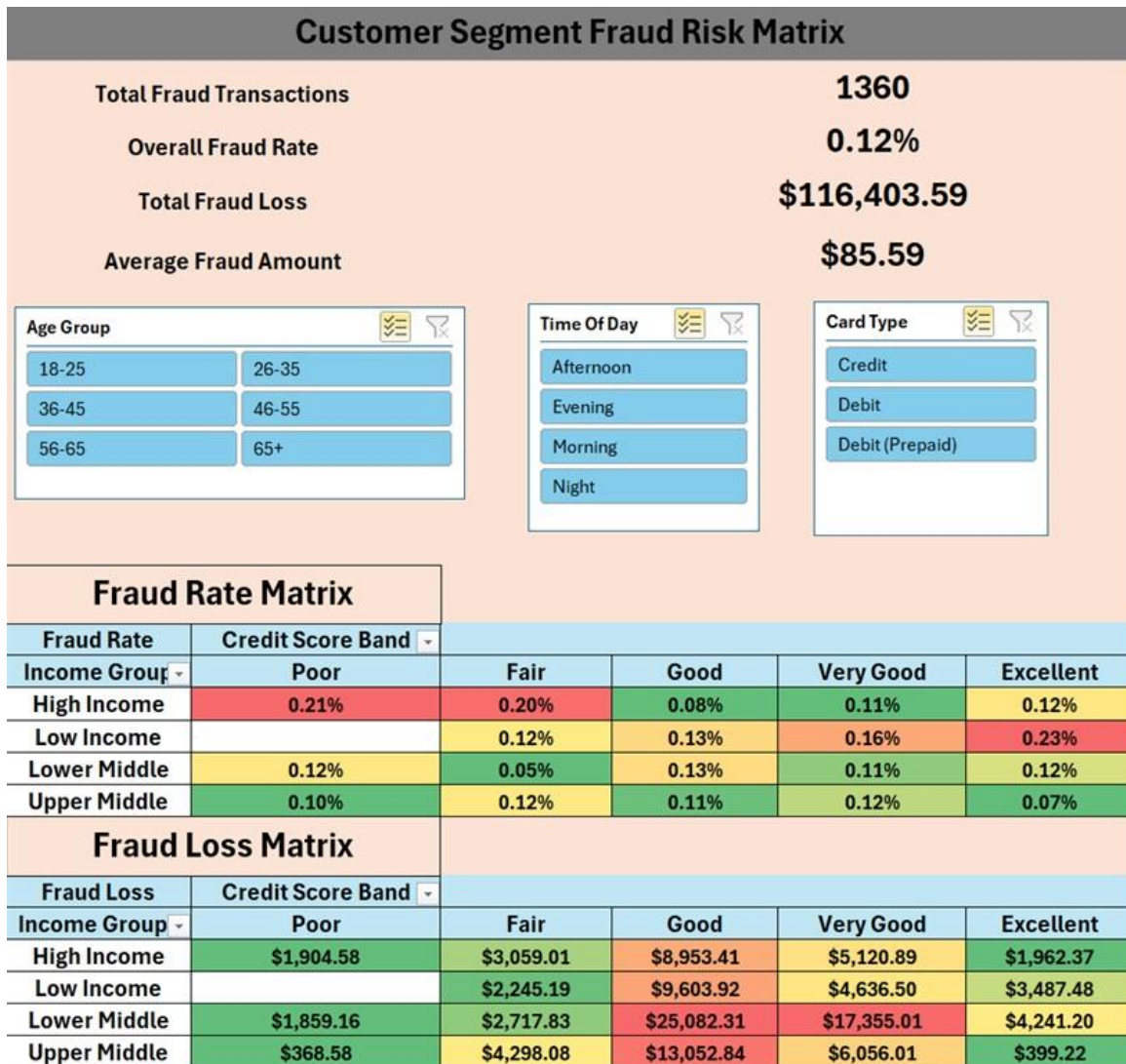
```
Total Spend := SUM (fact_transactions[amount])  
Total Transactions := COUNTROWS (fact_transactions)  
Average Transaction Amount := DIVIDE ([Total Spend], [Total Transactions])  
Fraud Transactions := CALCULATE (COUNTROWS (fact_transactions),  
fact_transactions [fraud_flag] =1)  
Fraud Loss := CALCULATE (SUM (fact_transactions[amount]),  
fact_transactions[fraud_flag] =1)  
Fraud Rate := DIVIDE ([Fraud Transactions], [Total Transactions])
```

4.6.1 Customer Segment Fraud Risk Analysis

Objective

Assess fraud exposure across customer segments by analyzing fraud rates and fraud losses across combinations of income groups and credit score bands. The analysis aims to identify customer profiles that exhibit elevated fraud risk and financial impact.

Customer Segment Fraud Risk Matrix



Key Insights

- Fraud exposure varied significantly across customer segments, indicating that fraud risk is not uniformly distributed across the customer base.
- Customers with **Poor** and **Fair** credit scores generally exhibited higher fraud rates than customers with stronger credit profiles, particularly within higher income segments.
- The **High Income** segment recorded some of the highest fraud rates in the analysis, with fraud rates exceeding 0.20% for customers in the Poor and Fair credit score bands.

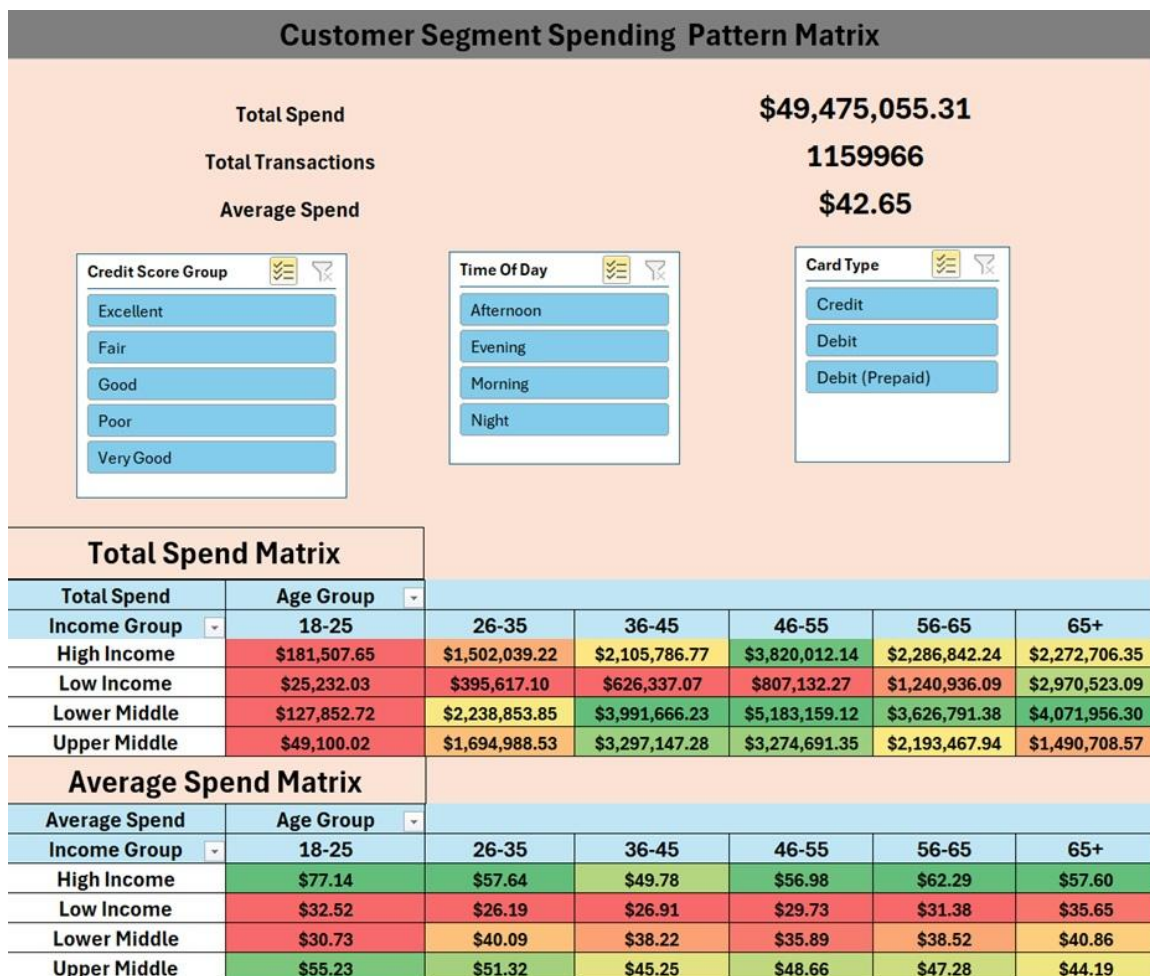
- Customers with **Good** credit scores generated the largest fraud losses despite not having the highest fraud rates, demonstrating the importance of evaluating both fraud frequency and monetary impact.
- Interactive slicers allow fraud patterns to be further analyzed by age group, card type, and time of day, supporting deeper investigation of segment-specific fraud behavior.

4.6.2 Customer Segment Spending Analysis

Objective

Analyze spending behavior across different customer income and age segments to identify the demographic groups contributing the highest transaction volume and spending value. The analysis also evaluates differences in average transaction value across customer segments to support customer profiling and targeted marketing strategies.

Customer Segment Spending Matrix



Key Insights

- Customers aged **46–55 years** generated the highest total spend across most income segments, indicating that middle-aged customers represent the most valuable spending cohort.
- **Lower Middle Income** customers contributed the largest share of overall spending, with peak spending exceeding **\$5.18M** in the 46–55 age group, suggesting that spending volume is not exclusively concentrated among high-income customers.
- **High Income** customers exhibited the highest average transaction values across nearly all age groups, with average spend ranging from **\$50–\$77 per transaction**, indicating stronger purchasing power despite lower overall transaction volume.
- **Low Income** customers consistently recorded the lowest average transaction values, typically between **\$26–\$36 per transaction**, reflecting more conservative spending behavior.
- Interactive slicers enabled further analysis of spending behavior by **Card Type**, **Credit Score Band**, and **Time of Day**, allowing deeper exploration of transaction trends across customer segments.

6.0 Power BI Analysis and Dashboards

While the Excel and Power Pivot analyses provided valuable insights into customer spending behavior and fraud patterns, Power BI was utilized to create a more interactive and visually driven analytical experience. A dimensional data model was built using the cleaned transaction fact table and supporting dimension tables for customers, cards, merchant categories, and dates. This enabled advanced filtering, drill-through navigation, time intelligence calculations, and dynamic cross-dimensional analysis.

Six interconnected dashboards were developed to examine customer spending behavior, customer segmentation, merchant performance, fraud exposure, and individual customer profiles. Together, these dashboards provide both executive-level summaries and detailed investigative capabilities for exploring transaction activity and fraud risk across the 2019 dataset.

6.1 Tools & Techniques Used

- Data Modeling & Star Schema Design
- DAX Measures & Time Intelligence Functions
- Interactive Slicers and Cross-Filtering
- Drill-Through Navigation
- Drill-Down Hierarchies
- Conditional Formatting
- Heatmaps
- KPI Cards
- Scatter Plot Analysis
- Treemaps
- Dynamic Tooltips
- Visual Hierarchy & Dashboard Design Principles

6.2 DAX Measures Created

The following DAX measures were created for specific tasks:

Date Table Creation

```
dim_date = CALENDAR (MIN (fact_transactions[date]),  
                      MAX (fact_transactions[date]))  
  
Month Name = FORMAT (dim_date [Date], "MMM")  
  
Month Number = MONTH (dim_date [Date])  
  
Quarter = "Q" & FORMAT (dim_date [Date], "Q")  
  
Weekday = FORMAT (dim_date [Date], "ddd")  
  
Year = YEAR (dim_date [Date])
```

Spending Measures Created

```

Total Spend = SUM (fact_transactions[amount])

Total Transactions = COUNTROWS (fact_transactions)

Average Spend = DIVIDE ([Total Spend], [Unique Customers])

Avg Transaction Value = DIVIDE ([Total Spend], [Total Transactions])

Previous Month Spend = CALCULATE ([Total Spend],
                                     DATEADD (dim_date [Date], -1, MONTH))

MoM Spend Growth % = DIVIDE ([Total Spend] - [Previous Month Spend],
                                [Previous Month Spend])

Customer Spend Rank = RANKX (ALL (dim_users[client_id]), [Total Spend],
                                , DESC, DENSE)

```

Fraud Measures Created

```

Fraud Transactions = CALCULATE (COUNTROWS (fact_transactions),
                                   fact_transactions[fraud_flag] = 1)

Fraud Loss = CALCULATE ([Total Spend], fact_transactions[fraud_flag] = 1)

Fraud Rate = DIVIDE ([Fraud Transactions], [Total Transactions])

Avg Fraud Amount = DIVIDE ([Fraud Loss], [Fraud Transactions])

MoM Fraud Rate Change (pp) = VAR PrevMonthRate = [Previous Month Fraud Rate]
                                RETURN
                                IF(ISBLANK(PrevMonthRate),
                                    BLANK (),
                                    ([Fraud Rate] - PrevMonthRate) * 100)

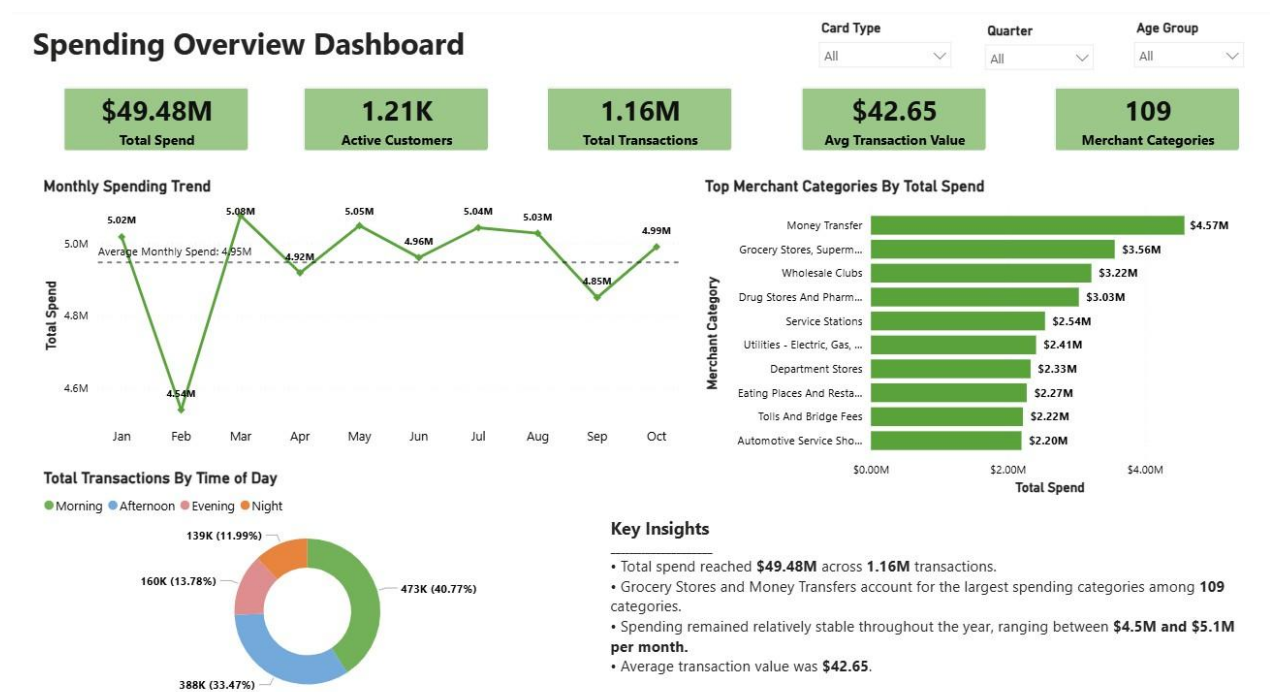
```

6.3 Spending Overview Dashboard

Objective

Provide an executive-level overview of transaction activity, spending trends, customer activity, and merchant category performance. The dashboard serves as the primary entry point for understanding overall spending behavior across the customer base.

Dashboard Image



Key Insights

- The platform processed **1.16 million transactions** totaling **\$49.48M** in customer spending during 2019, with an average transaction value of **\$42.65** across **1,210 unique customers**.
- Spending activity remained consistent throughout the year, with monthly totals ranging from approximately **\$4.5M** to **\$5.1M**, suggesting limited seasonality in transaction behavior.
- Money Transfer** represented the highest-spending merchant category at approximately **\$4.5M**, significantly exceeding most other categories.

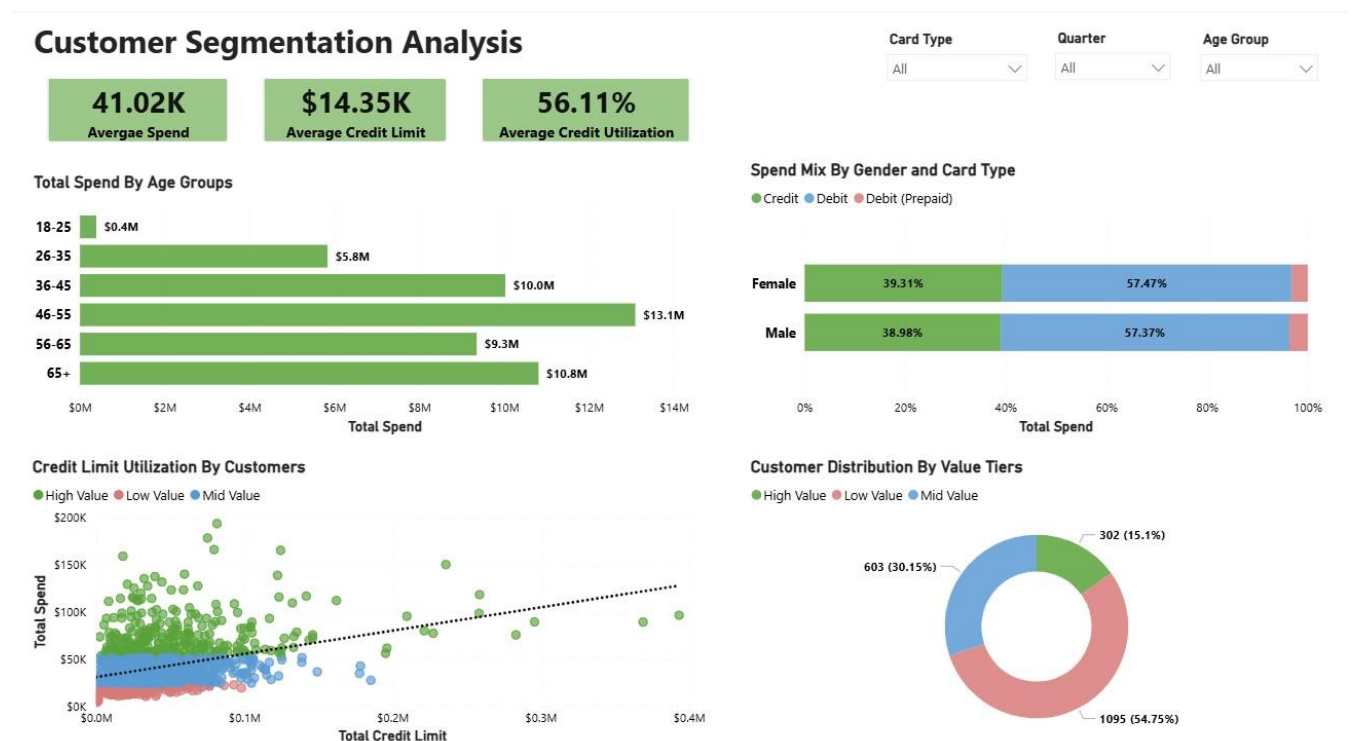
- Essential spending categories such as **Grocery Stores**, **Drug Stores**, and **Utilities** ranked among the highest contributors to total spending, reflecting everyday consumer spending behavior.
- Interactive analysis by **Age Group**, **Card Type**, and **Quarter** revealed meaningful differences in spending patterns, enabling deeper customer segmentation and behavioral analysis.

6.4 Customer Segmentation Dashboard

Objective

Segment customers based on demographic and financial characteristics to identify high-value customer groups, understand spending behavior, and evaluate the relationship between spending activity and credit utilization.

Dashboard Image



Key Insights

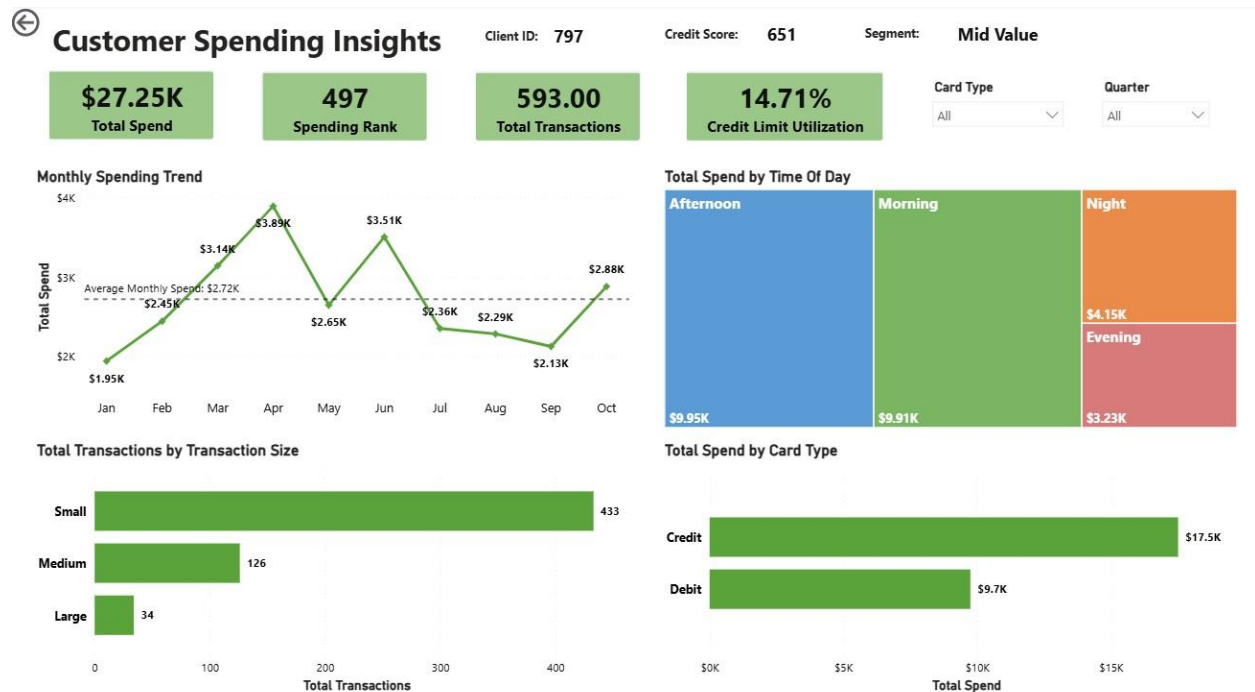
- Customers aged **46–55** generated the highest total spend (**\$13.1M**), followed by the **65+** (**\$10.8M**) and **36–45** (**\$10.0M**) age groups, indicating that spending activity peaks in middle-to-late adulthood.
- The customer base was dominated by **Low Value customers (54.8%)**, while **High Value customers represented only 15.1%** of customers, highlighting a highly concentrated customer value distribution.
- Across both male and female customers, **Debit cards accounted for approximately 57% of spending**, while Credit cards contributed around **39%**, suggesting a strong preference for debit-based transactions.
- The credit utilization scatter plot shows a **positive relationship between available credit and spending**, with customers holding larger credit limits generally exhibiting higher spending levels.
- High Value customers consistently occupied the upper range of both **credit limits and spending amounts**, indicating that customer value is strongly associated with purchasing capacity and transaction activity.

6.5 Customer Insights Dashboard (Drill-Through)

Objective

Provide a customer-level analytical view of spending behavior, transaction patterns, and credit utilization to support customer profiling, identify key spending drivers, and enable detailed investigation of individual customer activity through drill-through analysis.

Dashboard Image



Key Insights

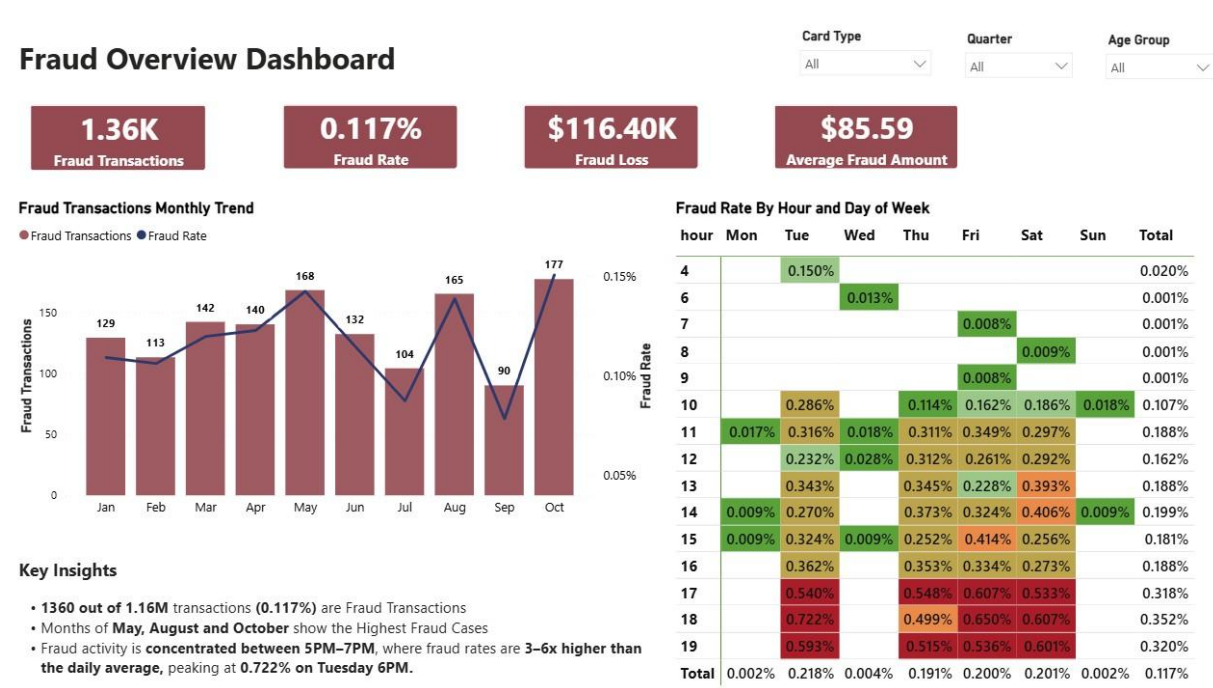
- Customer with Client ID 797 generated \$27.25K in total spend across 593 transactions, placing them 497th in the overall spending ranking and classifying them as a Mid Value customer.
- The customer's spending was concentrated in Credit Card transactions (\$17.5K, 64%), compared with Debit Card transactions (\$9.7K, 36%), indicating a strong preference for credit-based spending.
- Spending activity was relatively consistent throughout the year, with an average monthly spend of \$2.72K. The highest spending months were April (\$3.89K) and June (\$3.51K), while January (\$1.95K) and September (\$2.13K) recorded the lowest spending levels.
- Transaction behavior was dominated by small-value purchases, which accounted for 433 of 593 transactions (73%), compared with 126 medium-value and 34 large-value transactions, suggesting frequent low-ticket spending activity.
- Spending was concentrated during Afternoon (\$9.95K) and Morning (\$9.91K) periods, which together represented approximately 73% of total spend, while Evening and Night transactions contributed a substantially smaller share.

6.5 Fraud Overview Dashboard

Objective

Monitor and quantify fraud activity across the transaction ecosystem by tracking fraud volume, financial losses, temporal patterns, and high-risk transaction periods to support fraud detection and risk mitigation efforts.

Dashboard Image



Key Insights

- Out of **1.16 million transactions**, only **1,360 transactions** were identified as fraudulent, resulting in an overall **fraud rate of 0.117%** and a total estimated **fraud loss of \$116.4K**.
- Fraud activity was not evenly distributed throughout the year. **October (177 cases), May (168 cases), and August (165 cases)** recorded the highest numbers of fraudulent transactions, indicating periodic spikes in fraud occurrence.

- The average fraud event resulted in a loss of **\$85.59**, suggesting that overall fraud exposure is driven more by the frequency of incidents than by exceptionally large individual fraud amounts.
- Fraud was heavily concentrated during the **late afternoon and evening hours (5 PM–7 PM)**, where fraud rates consistently exceeded **0.5%**, significantly above the overall average fraud rate of **0.117%**.
- The single highest-risk period occurred on **Tuesday at 6 PM**, where the fraud rate reached **0.722%**, more than **6 times higher than the overall fraud rate**, highlighting a clear temporal hotspot for fraudulent activity.

6.6 Fraud Analysis Dashboard

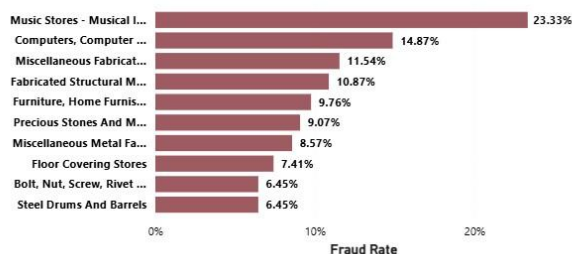
Objective

Identify the key factors associated with fraudulent activity by analyzing fraud rates and fraud losses across merchant categories, card types, transaction sizes, and transaction timing, enabling targeted fraud prevention and risk management strategies.

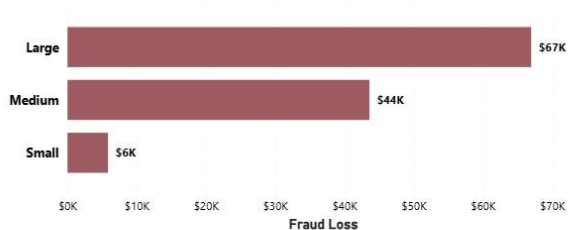
Dashboard Image

Fraud Analysis -Factors Affecting Fraud

Highest Fraud Rates By Merchant Category



Fraud Loss By Transaction Size

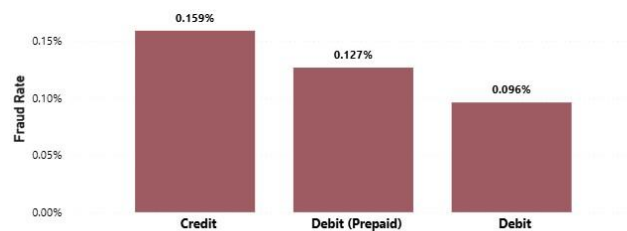


Key Insights

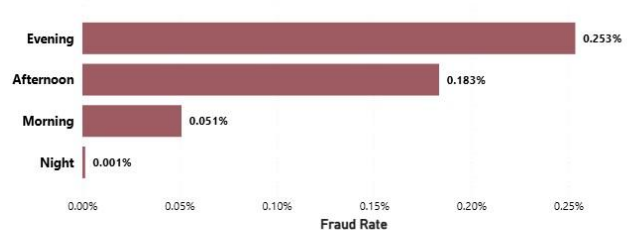
- Credit Cards shows the highest fraud rate of ~ 0.16% .
- Large transactions account for 67K (57.52%) of total fraud loss.

Card Type: All
Quarter: All
Age Group: All

Fraud Rate by Card Type



Fraud Rate By Time of Day



- Evening Time is associated with the highest risk of Fraud with 0.25% fraud rate.
- Music Stores and Computers are the Highest risk categories with fraud Rates of 23.33% and 14.87% respectively.

Key Insights

- **Merchant category risk varies substantially across sectors.** *Music Stores – Musical Instruments* recorded the highest fraud rate at **23.33%**, followed by *Computers & Computer Peripheral Equipment* (**14.87%**) and *Miscellaneous Fabricated Metal Products* (**11.54%**), indicating that fraud exposure is highly concentrated in specific merchant categories.
- **Credit cards exhibited the highest fraud rate among payment methods.** Credit card transactions recorded a fraud rate of **0.159%**, compared with **0.127%** for Debit (Prepaid) cards and **0.096%** for standard Debit cards, making credit card activity approximately **66% riskier than debit transactions**.
- **Large-value transactions accounted for most financial fraud losses.** Although less frequent than smaller transactions, large transactions generated approximately **\$67K in fraud losses**, representing **57.5% of total fraud loss**, compared with **\$44K** from medium transactions and only **\$6K** from small transactions.
- **Fraud risk increased significantly during evening hours.** Transactions occurring in the **Evening** recorded the highest fraud rate (**0.253%**), followed by the **Afternoon (0.183%)**, while **Night transactions (0.001%)** showed negligible fraud activity.
- **Fraud exposure is driven by a combination of transaction context and customer behavior.** High-risk merchant categories, credit card usage, larger transaction values, and evening transaction activity consistently emerged as the strongest indicators of elevated fraud risk across the dataset.

6.7 Merchant Insights Dashboard

Objective

Provide a merchant-level view of both spending performance and fraud exposure, enabling stakeholders to identify high-value merchant categories, monitor transaction behavior, and assess fraud risk within individual merchant segments.

Dashboard Image



Key Insights

- The selected merchant category (**Grocery Stores & Supermarkets**) generated approximately **\$3.56M in spend across 139K transactions**, making it one of the highest-volume merchant segments in the portfolio. Despite this scale, the category maintained a relatively low **fraud rate of 0.09%**.
- Spending remained highly stable throughout the year, averaging **\$355.7K per month**. Monthly spend fluctuated within a narrow range of **\$329K–\$373K**, indicating consistent customer demand and limited seasonality.
- Morning and Afternoon transactions accounted for nearly 77% of total spend**, contributing **\$1.41M** and **\$1.33M** respectively. Evening and Night transactions represented a much smaller share of merchant revenue.
- Although Morning transactions generated the highest spending volume, **Afternoon transactions produced the largest fraud losses (\$2.65K)**, followed by Evening transactions (**\$1.71K**). This suggests fraud risk is not perfectly aligned with transaction volume.

- Fraud rates remained relatively low throughout the year but showed periodic spikes. The highest monthly fraud rates occurred in **April, May, and August (0.12%)**, approximately **three times higher than March (0.04%)**, indicating specific periods that may warrant closer monitoring.

7.0 Summary of Insights

- **Customers aged 46–55 generated the highest total spend (\$13.1M)**, with spending generally increasing across older age groups and higher credit limits.
- **High Value customers represented only 15.1% of the customer base (302 customers)** but contributed a disproportionately large share of overall spending and exhibited the highest credit utilization levels.
- **Debit cards accounted for approximately 57% of total spending** for both male and female customers, indicating highly consistent payment behavior across genders.
- Despite processing **1.16 million transactions worth \$49.5M**, only **1,360 transactions were identified as fraudulent**, resulting in a relatively low **overall fraud rate of 0.117%** and **total fraud losses of \$116.4K**.
- Fraud risk was **not evenly distributed across the transaction portfolio**, with certain merchant categories exhibiting significantly elevated fraud exposure.
- **Music Stores (23.33% fraud rate)** and **Computer Equipment Stores (14.87% fraud rate)** recorded the highest fraud rates, substantially exceeding the portfolio-wide fraud average.
- **Credit card transactions experienced the highest fraud rate (0.159%)**, compared with Debit and Debit (Prepaid) transactions.
- **Large-value transactions accounted for approximately 57.5% of total fraud losses (\$67K)**, highlighting the disproportionate financial impact of high-value fraud events.
- Fraud activity was strongly concentrated within specific time periods, with **Evening transactions recording the highest fraud rate (0.253%)**.
- The single highest fraud hotspot occurred on Tuesday **6 PM**, where the fraud rate reached **0.722%**, more than **six times higher than the overall fraud rate**, indicating that transaction timing is a critical factor in fraud monitoring and detection.

8.0 Business Recommendations

- **Prioritize monitoring of high-risk merchant categories** by implementing enhanced fraud controls and transaction reviews for categories exhibiting unusually high fraud rates.
- **Apply risk-based fraud screening for large transactions**, as large-value purchases contribute majority of the fraud-related financial losses despite representing a smaller share of transaction volume.
- **Increase fraud surveillance during peak-risk periods**, particularly during evening hours and identified weekday fraud hotspots, to improve detection effectiveness and reduce losses.
- **Develop targeted retention strategies for High Value customers**, who represent a relatively small portion of the customer base but contribute a disproportionate share of overall spending.
- **Leverage customer segmentation for personalized product offerings**, focusing on age groups and customer tiers that demonstrate the strongest spending behavior and credit utilization.
- **Implement dynamic fraud risk scoring models** that combine merchant category, transaction amount, payment method, and transaction timing to identify potentially fraudulent activity more accurately than rule-based approaches.

9.0 Conclusion

This project demonstrates a **complete analytics workflow** spanning data preparation in Python, exploratory analysis in Excel and Power Pivot, and interactive dashboard development in Power BI.

By transforming over **13 million raw transaction records** into a structured analytical model, the analysis uncovered **key drivers of customer spending** behavior and **fraud risk**. Distinct customer segments, spending patterns, high-risk merchant categories, and fraud hotspots were identified, **providing insights** that would not be apparent from raw transactional data alone.

The resulting analytical assets—including **Excel investigation tools**, **Power Pivot models**, and six interactive **Power BI dashboards**—deliver a comprehensive framework for customer segmentation, spending analysis, fraud monitoring, and risk assessment.